ASSIGNMENT - 4

Problems on Fourier series:

Obtain the Fourier Series for the following:

1. $f(x) = \begin{cases} -1 & -\pi < t < -\pi/2 \\ 0 & -\pi/2 < t < \pi/2 \\ 1 & \pi/2 < t\pi \end{cases}$ in $(-l, l)$

   Ans: $f(x) = \frac{2}{\pi}\left[\!\left[sint - sin2t + \frac{1}{3}sin3t + \cdots \right]\!\right]$

2. $f(x) = |sinx| \quad -\pi < x < \pi$

   Ans: $\frac{2}{\pi} - \frac{4}{\pi}\left[\!\left[\frac{cos2x}{3} + \frac{cos4x}{15} + \quad \right]\!\right]$